What is deep copy :

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* It is a technique through which we can create multiple objects.
* We can copy the data of one object to another object.
* Here two different objects are created so, if we modify first object data then second object data will remain unchanged.

```java
package com.ravi.deep_copy;

public class EmployeeDemo
{
    public static void main(String[] args)
    {
        IO.println("Before Copy");
        Employee e1 = new Employee(); //id = 0   name = null
        Employee e2 = new Employee(111, "John"); //id = 111   name = John
        IO.println(e1+" : "+e2);

        IO.println("After Copy :");
        e1.setId(e2.getId()); //Copy the 2nd object data to 1st Object
        e1.setName(e2.getName());
        IO.println(e1+" : "+e2);

        IO.println("Modify First Object Data i.e. e1 :");
        e1.setId(999);
        e1.setName("Raj");
        IO.println(e1+" : "+e2);

    }
}

class Employee
{
    private int id;
    private String name;

    public Employee()  //NO Argument constructor
    {
        id = 0;
        name = null;
    }


    public Employee(int id, String name) //Parameterized constructor
    {
        super();
        this.id = id;
        this.name = name;
    }


    public int getId()
    {
        return id;
    }


    public void setId(int id)
    {
        this.id = id;
    }


    public String getName()
    {
        return name;
    }


    public void setName(String name)
    {
        this.name = name;
    }


    @Override
    public String toString()
    {
        return "Employee [id=" + id + ", name=" + name + "]";
    }

}
```

***** Pass By Value :**

Java does not support **pointers** so java only works with pass by value.

Pass by value means we are sending the **copy of original data** to the method.

For primitive types : The actual value is copied.

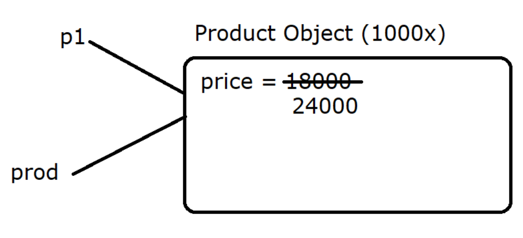
For objects : The reference (address) is copied, not the actual object. So both references still point to the same object in memory.

Example of Pass By Value :

```java
public class PassByValueDemo1
{
    public static void main(String[] args)
    {
        int x = 10;
        accept(x);
        IO.println(x);
    }

    public static void accept(int y)  //y = 10
    {
        y = y + 20;
    }
}
```

Diagram for PassByValueDemo2.java




```java
package com.ravi.pass_by_value;

public class PassByValueDemo2
{
    public static void main(String[] args)
    {
        Product p1 = new Product(18000);
        IO.println("Before Method Call :"+p1.getPrice()); //18000
        accpet(p1);
        IO.println("After Method Call :"+p1.getPrice()); //24000

    }

    public static void accpet(Product prod)
    {
        prod.setPrice(24000);
    }
}

class Product
{
    private double price;

    public Product(double price)
    {
        this.price = price;
    }

    public double getPrice()
    {
        return price;
    }

    public void setPrice(double price)
    {
        this.price = price;
    }

}
```

Assignment :

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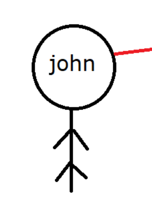
```java
class Customer
{
  private int id;
  private String name;
  private double balance;
  private int creditScore;
  private String address;

        //Methods

}
```




```java
//HiTech City
public class LoanDept
{
  public void getEligibility(Customer cust)
  {
    int cs = cust.getCreditScore();

    if(cs >=750)
    {
     IO.println("Eligible for Home Loan");
    }

  }

}
```

```java
package com.ravi.pass_by_value;

public class PassByValueDemo3
{
    public static void main(String[] args)
    {
       Student s1 = new Student(111, "Raj");
       Student s2 = new Student(222, "Priya");
       Student s3 = new Student(333, "Elina");

       College clg = new College();
       clg.getStudentInformation(s3);
    }
}

class Student
{
    private int id;
    private String name;

    public Student(int id, String name)
    {
        super();
        this.id = id;
        this.name = name;
    }

    public int getId() {
        return id;
    }

    public String getName() {
        return name;
    }
}


class College
{
    public void getStudentInformation(Student stud) //stud = s1
    {
        IO.println("Getting Student Information in the College");
        IO.println("Student id is :"+stud.getId());
        IO.println("Student Name is :"+stud.getName());

    }
}
```

```java
package com.ravi.pass_by_value;

public class PassBuValueDemo4 {

    public static void main(String[] args)
    {
        Employee alen = new Employee(101, "Mr Alen", 50000);
        IO.println("Original Data"+alen);

        ITCompany company = new ITCompany();
        company.getEmployment(alen);
    }
}

class Employee
{
    private int id;
    private String name;
    private double salary;

    public Employee(int id, String name, double salary)
    {
        super();
        this.id = id;
        this.name = name;
        this.salary = salary;
    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public double getSalary() {
        return salary;
    }

    public void setSalary(double salary) {
        this.salary = salary;
    }

    @Override
    public String toString()
    {
        return "Employee [id=" + id + ", name=" + name + ", salary=" + salary + "]";
    }

}

class ITCompany
{
    public void getEmployment(Employee e1)
    {
        IO.println("After Joining new Company");

        IO.println("Employee Id is :"+e1.getId());
        IO.println("Employee Name is :"+e1.getName());
        e1.setSalary(e1.getSalary() + 25000);
        IO.println("Employee Salary is :"+e1.getSalary());


    }
}
```